

EXAMINATION PAPER: ADVANCED ALGORITHMS

Section A: Theory

1. Define asymptotic notation and explain Big-O notation.

- (A) Upper bound on running time
- (B) Lower bound on running time
- (C) Exact bound on running time
- (D) None of the above

Ans: A

2. In a distributed consensus protocol running across multiple unreliable network partitions, which of the following

- (A) Consistency and Latency
- (B) Consistency and Availability
- (C) Partition Tolerance and Durability
- (D) Atomicity and Isolation

Ans: B

3. Which sorting algorithm exhibits the best worst-case performance among comparison sorts?

- (A) Quick Sort
- (B) Merge Sort
- (C) Bubble Sort
- (D) Insertion Sort

Ans: B